

NETWORKWALKS

Week 1 Networking Lab Report

Setting up a Virtual Lab with VirtualBox and Kali Linux

Submitted by

Rishu Satish Singh

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2. Objective

The objective of Week 1 of the Networkwalks Networking Lab is to build a working virtual lab environment that will be used for the remaining weeks of the course. This includes:

- Installing and configuring Oracle VirtualBox on a Windows host machine.
- Setting up Kali Linux as a guest virtual machine inside VirtualBox.
- Configuring the virtual machine's network adapter and understanding VirtualBox's NAT Network feature.
- Verifying the IP address obtained by the Kali Linux guest and confirming that basic networking is functional.
- Testing internet and DNS connectivity from inside the Kali Linux virtual machine.
- Configuring host-guest integration features such as shared clipboard, drag-and-drop, and shared folders.
- Recording any problems faced during setup and the steps taken to resolve them.

3. Tools and Software Used

Item	Details
Host operating system	Windows
Virtualization software	Oracle VirtualBox Manager
Guest operating system	Kali Linux
Virtual machine name	kali
Guest disk / memory	25.00 GB virtual disk (kali.vdi), 2048 MB base memory
Networking tools used	Kali Linux terminal commands: ip a, ping
Documentation tool	Screenshots captured directly from the host and guest desktop

4. VirtualBox and Kali Linux Installation / Setup

Oracle VirtualBox was installed on the Windows host machine, and a new virtual machine named “kali” was created to run Kali Linux as the guest operating system. The VirtualBox Manager overview for this machine is shown in Figure 1.

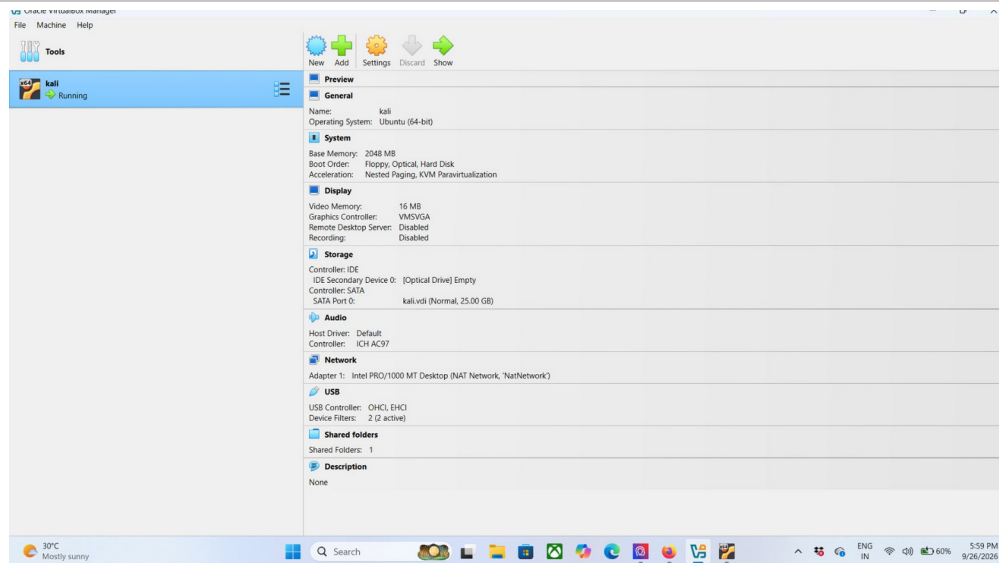


Figure 1: VirtualBox Manager overview of the “kali” virtual machine, showing its running state and configuration summary.

Figure 1 confirms that the virtual machine is named “kali”, is currently in the Running state, uses Ubuntu (64-bit) as the VirtualBox-assigned OS type, has 2048 MB of base memory, and stores its data on a 25.00 GB virtual disk (kali.vdi) attached to the SATA controller. The boot order is set to Floppy, Optical, Hard Disk, and hardware acceleration (Nested Paging, KVM Paravirtualization) is enabled. One shared folder is also listed, which is covered in Section 8.

Additional general settings for the virtual machine are shown in Figure 2, covering the Advanced tab (snapshot folder, shared clipboard and drag-and-drop modes) and the System / Motherboard tab (base memory allocation and boot order).

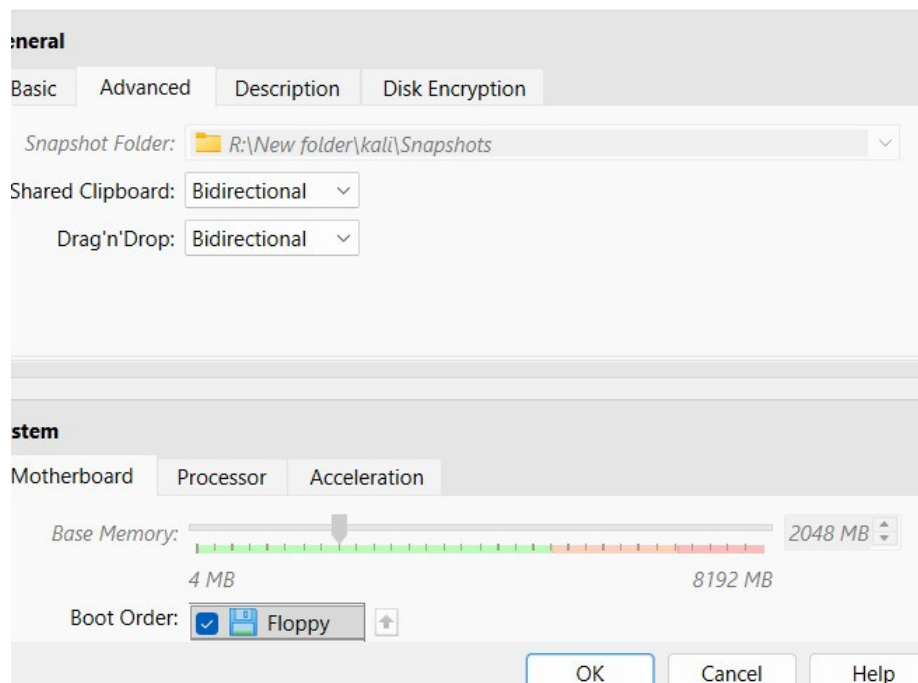


Figure 2: VirtualBox “kali” machine settings — General/Advanced tab and System/Motherboard tab.

Figure 2 shows the snapshot folder path, the Shared Clipboard and Drag'n'Drop modes (both set to Bidirectional), and confirms the base memory slider is set to 2048 MB, with Floppy selected first in the boot order. These clipboard and drag-and-drop settings are discussed further in Section 8.

5. Network Adapter Configuration

The virtual machine's network adapter was configured to use VirtualBox's NAT Network feature rather than the default NAT engine. Figure 3 shows the Network settings page for Adapter 1 of the “kali” machine.

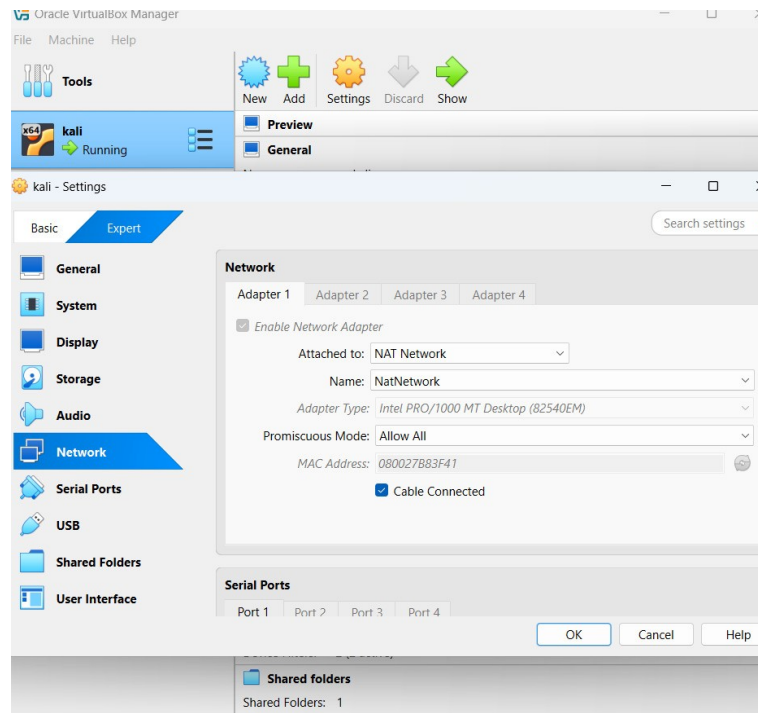


Figure 3: Network settings for Adapter 1, attached to NAT Network “NatNetwork”.

Figure 3 confirms that Adapter 1 is enabled, attached to “NAT Network”, using the network named “NatNetwork”, with adapter type Intel PRO/1000 MT Desktop (82540EM), Promiscuous Mode set to “Allow All”, MAC address 08:00:27:B8:3F:41, and the cable marked as connected.

The NAT Network object itself (“NatNetwork”) was configured through VirtualBox's global Network settings, shown in Figure 4.

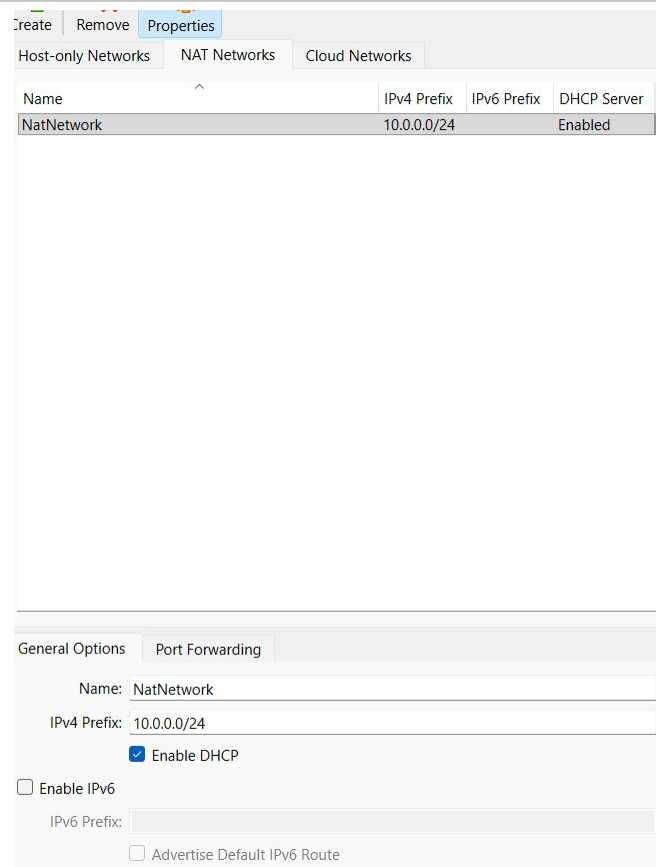
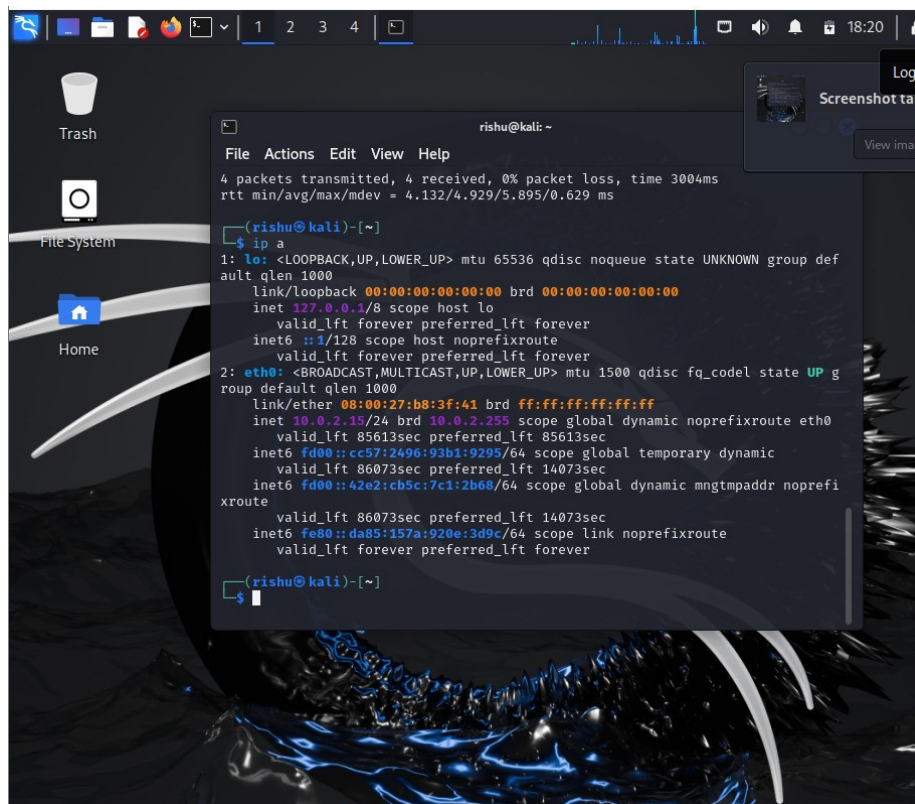


Figure 4: NAT Network “NatNetwork” properties, showing the IPv4 prefix and DHCP setting.

Figure 4 shows that the network named “NatNetwork” uses the IPv4 prefix 10.0.0.0/24, with the DHCP server enabled and IPv6 disabled. This means VirtualBox is configured to hand out addresses in the 10.0.0.0/24 range with a default gateway of 10.0.0.1 to any virtual machine attached to this network. Whether the Kali guest actually received an address in this range is checked in Section 6.

6. Kali Linux IP Configuration and Verification

Inside the Kali Linux guest, the command `ip a` was run in a terminal to display the current network interface configuration. The output is shown in Figure 5.



```

File Actions Edit View Help
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 4.132/4.929/5.895/0.629 ms

(rishu@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
  aut qlen 1000
  link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
  inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
  inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
  roup default qlen 1000
  link/ether 08:00:27:b8:3f:41 brd ff:ff:ff:ff:ff:ff
  inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
    valid_lft 85613sec preferred_lft 85613sec
  inet6 fd00::cc57:2496:93b1:9295/64 scope global temporary dynamic
    valid_lft 86073sec preferred_lft 14073sec
  inet6 fd00::42e2:cb5c:7c1:2b68/64 scope global dynamic mngtmpaddr noprefi
    xroute
    valid_lft 86073sec preferred_lft 14073sec
  inet6 fe80::da85:157a:920e:3d9c/64 scope link noprefixroute
    valid_lft forever preferred_lft forever

(rishu@kali)-[~]
$

```

Figure 5: Output of “ip a” inside Kali Linux, showing the eth0 interface configuration.

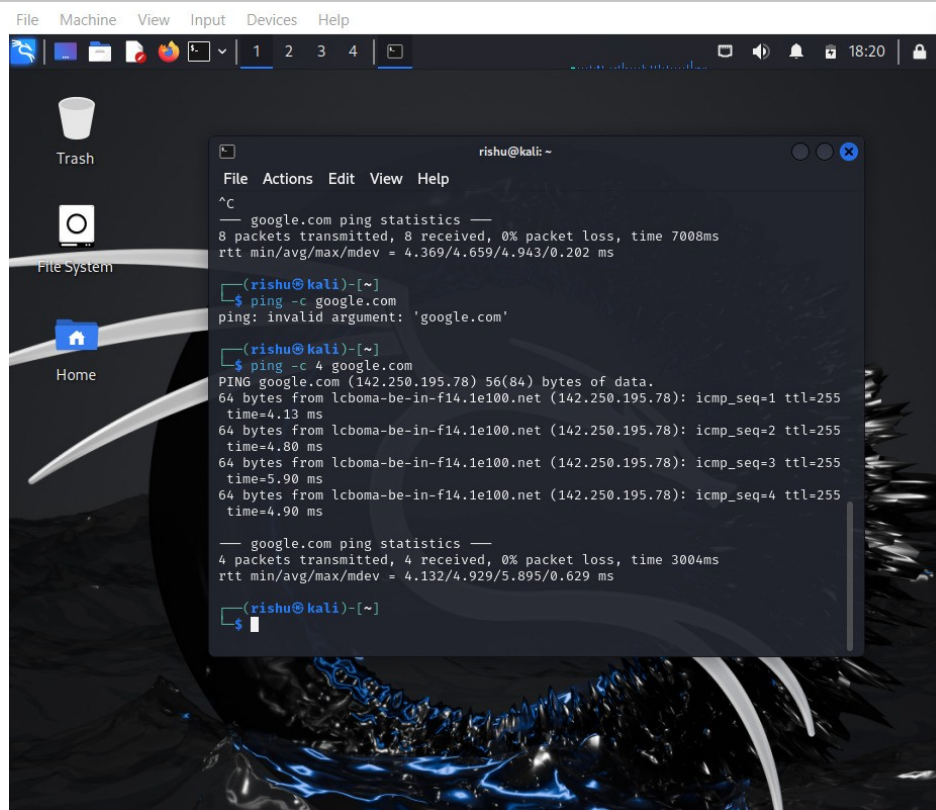
Figure 5 shows that the eth0 interface is UP, with MAC address 08:00:27:b8:3f:41 — this matches the MAC address of Adapter 1 shown in Figure 3, confirming that this is the correct interface. The IPv4 address actually assigned to eth0 is 10.0.2.15/24 (broadcast 10.0.2.255), obtained dynamically. Several IPv6 addresses are also listed on the same interface.

Observation: the address actually obtained (10.0.2.15/24) is from VirtualBox's default NAT range, not from the 10.0.0.0/24 range configured for “NatNetwork” in Figure 4. Even though the adapter is set to attach to the NAT Network named “NatNetwork” (Figure 3), the address handed out does not match that network's configured prefix. This mismatch is treated as an open issue and is discussed in Section 9.

Pending: a static IP address of 10.0.0.2/24 has not yet been configured on the Kali guest. No screenshot currently shows a static IP assignment, so this task is marked as not completed and will need to be carried out and documented in a follow-up screenshot.

7. Internet and DNS Connectivity Testing

Connectivity from the Kali Linux guest was tested using the ping command. Figure 6 shows the terminal output of this test.

The image shows a Kali Linux desktop environment. In the background, there are icons for 'Trash', 'File System', and 'Home'. A terminal window is open in the foreground, displaying the following text:

```
rishu@kali: ~  
File Actions Edit View Help  
^C  
— google.com ping statistics —  
8 packets transmitted, 8 received, 0% packet loss, time 7008ms  
rtt min/avg/max/mdev = 4.369/4.659/4.943/0.202 ms  
  
(rishu@kali)-[~]  
$ ping -c google.com  
ping: invalid argument: 'google.com'  
  
(rishu@kali)-[~]  
$ ping -c 4 google.com  
PING google.com (142.250.195.78) 56(84) bytes of data:  
64 bytes from lcboma-be-in-f14.1e100.net (142.250.195.78): icmp_seq=1 ttl=255  
time=4.13 ms  
64 bytes from lcboma-be-in-f14.1e100.net (142.250.195.78): icmp_seq=2 ttl=255  
time=4.80 ms  
64 bytes from lcboma-be-in-f14.1e100.net (142.250.195.78): icmp_seq=3 ttl=255  
time=5.90 ms  
64 bytes from lcboma-be-in-f14.1e100.net (142.250.195.78): icmp_seq=4 ttl=255  
time=4.90 ms  
  
— google.com ping statistics —  
4 packets transmitted, 4 received, 0% packet loss, time 3004ms  
rtt min/avg/max/mdev = 4.132/4.929/5.895/0.629 ms  
  
(rishu@kali)-[~]  
$
```

Figure 6: Ping test from Kali Linux to google.com.

Figure 6 shows that an initial attempt (ping google.com) failed with an “invalid argument” error because no packet count flag was supplied. The corrected command, ping -c 4 google.com, succeeded: the hostname google.com was resolved to the IP address 142.250.195.78, and all 4 packets were transmitted and received with 0% packet loss (round-trip time between 4.132 ms and 5.895 ms).

Because this test used a hostname (google.com) rather than a raw IP address, a successful result demonstrates two things at once: (1) the DNS resolver configured on the Kali guest was able to resolve the domain name to an IP address, and (2) the guest was able to reach that IP address over the internet.

Pending: a separate ping test to a raw IP address such as 8.8.8.8 (to check internet reachability independently of DNS) was not captured in a screenshot. This is recorded as pending and should be captured separately for completeness.

8. Shared Clipboard, Drag-and-Drop, and Shared Folder Configuration

VirtualBox provides several host-guest integration features. Their configured status, as set in the “kali” machine settings, is described below.

8.1 Shared Clipboard and Drag-and-Drop

As shown earlier in Figure 2, the Shared Clipboard mode and the Drag'n'Drop mode are both set to Bidirectional in the General → Advanced settings of the “kali” virtual machine. This means the configuration allows copy-paste and drag-and-drop operations to work in both directions (host to guest and guest to host).

Completed: the Bidirectional configuration shown in Figure 2 was tested by the student, who confirmed that clipboard copy-paste and drag-and-drop both work correctly between the host and the Kali guest. This confirms that VirtualBox Guest Additions are installed and functioning correctly on the guest.

8.2 Shared Folder

A shared folder was configured for the “kali” virtual machine, as shown in Figure 7.

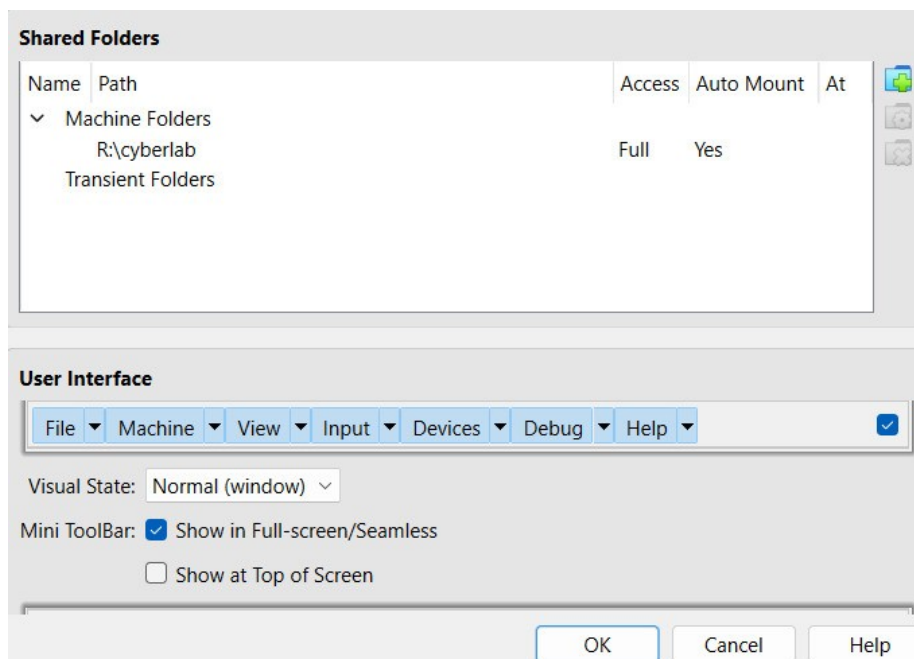


Figure 7: Shared Folders configuration and User Interface settings for the “kali” virtual machine.

Figure 7 shows one machine folder configured: the host path R:\cyberlab is shared with Full access and Auto Mount enabled. This matches the “Shared folders: 1” entry seen earlier in Figure 1.

Completed: the student confirmed that this shared folder is working and accessible from inside the Kali Linux file system, with Full access and Auto Mount enabled as configured in Figure 7.

9. Problems Encountered and Troubleshooting Performed

The table below lists the issues noticed while completing this lab, based only on what the screenshots show and what was reported during testing.

Problem	What was observed	Status / next step
Ping command syntax error	“ping google.com” returned “invalid argument” (Figure 6) because no count flag was given.	Resolved: re-ran as “ping -c 4 google.com”, which succeeded.
IP address does not match NAT Network prefix	“NatNetwork” is configured for 10.0.0.0/24 (Figure 4), but eth0 obtained 10.0.2.15/24 (Figure 5) — the default VirtualBox NAT range.	Open issue: needs adapter re-check, DHCP lease renewal, or VM restart, followed by a fresh “ip a” screenshot.
NAT Network gateway unreachable	Reported during testing that the NatNetwork gateway (10.0.0.1) could not be reached.	Not captured in a screenshot. Marked pending; a ping to 10.0.0.1 should be run and documented.
Static IP not yet configured	Task requires a static address of 10.0.0.2/24 on the Kali guest; no	Pending: configure static IP and capture verification screenshot.

Problem	What was observed	Status / next step
	configuration screenshot exists yet.	

10. Results and Conclusion

10.1 What was successfully demonstrated

- Oracle VirtualBox was installed on the Windows host, and a Kali Linux virtual machine named “kali” was created and is running (Figure 1).
- Adapter 1 of the Kali VM was attached to a NAT Network named “NatNetwork”, configured with prefix 10.0.0.0/24 and DHCP enabled (Figures 3 and 4).
- The Kali guest automatically obtained an IPv4 address on its eth0 interface, confirmed via the “ip a” command (Figure 5).
- Internet access and DNS resolution were both demonstrated together through a successful ping to google.com, which resolved to 142.250.195.78 with 0% packet loss (Figure 6).
- Shared Clipboard and Drag'n'Drop were configured to Bidirectional mode (Figure 2) and confirmed working by the student between the host and the Kali guest.
- A shared folder (R:\cyberlab) was configured with Full access and Auto Mount (Figure 7) and confirmed working and accessible by the student inside the Kali guest.

10.2 What remains pending

- The IP address obtained (10.0.2.15/24) does not match the configured NatNetwork prefix (10.0.0.0/24); this needs to be investigated and re-verified.
- A static IP address of 10.0.0.2/24 has not yet been configured on the Kali guest.
- The reported failure to reach the NatNetwork gateway (10.0.0.1) was not captured in a screenshot and needs to be re-tested with evidence.
- A ping test to a raw IP address (such as 8.8.8.8) has not been separately captured.

Overall, the core Week 1 setup goal — installing VirtualBox, creating a Kali Linux virtual machine, attaching it to a network, confirming basic internet and DNS connectivity, and enabling host-guest integration features — has been achieved. The remaining items relate specifically to the NAT Network addressing issue and the static IP task, which are clearly identified above and should be completed and documented before this lab is considered fully finished.

11. Screenshot Evidence — List of Figures

All screenshots used as evidence in this report are listed below with their figure numbers, captions, and what each one demonstrates. The full-size screenshots appear inline in the relevant sections above (Sections 4–8).

Figure	Caption	What it proves
Figure 1	VirtualBox Manager overview of the “kali” virtual machine	Confirms the VM exists, is running, and its base hardware configuration.
Figure 2	General/Advanced and System/Motherboard settings	Confirms base memory, boot order, and Bidirectional clipboard/drag-and-drop settings.

Figure	Caption	What it proves
Figure 3	Network Adapter 1 settings (NAT Network "NatNetwork")	Confirms the adapter is attached to NAT Network "NatNetwork" with cable connected.
Figure 4	NAT Network "NatNetwork" properties	Confirms the configured IPv4 prefix (10.0.0.0/24) and that DHCP is enabled.
Figure 5	"ip a" output inside Kali Linux	Confirms the actual IP address (10.0.2.15/24) obtained on eth0.
Figure 6	Ping test to google.com	Confirms successful DNS resolution and internet reachability (0% packet loss).
Figure 7	Shared Folders and User Interface settings	Confirms one shared folder (R:\cyberlab) is configured with Full access.

12. Week 1 Completion Checklist

This checklist summarizes the status of every task in this lab, based on the screenshots in this report together with the student's direct confirmation of features tested on the live system. Items marked "Pending" are the tasks that are genuinely not yet complete and still require configuration or testing.

Task	Status	Evidence
Install VirtualBox on Windows host	Completed	Figure 1
Create Kali Linux virtual machine	Completed	Figure 1
Configure base memory / boot order	Completed	Figure 1, Figure 2
Attach adapter to NAT Network ("NatNetwork")	Completed	Figure 3
Configure NAT Network prefix and DHCP	Completed	Figure 4
Obtain IP address on Kali guest (DHCP)	Completed (address range mismatch noted)	Figure 5
Configure static IP 10.0.0.2/24	Pending	Not captured
Verify NAT Network gateway (10.0.0.1) reachability	Pending	Not captured
Test internet + DNS via ping to google.com	Completed	Figure 6
Test internet via ping to raw IP (e.g. 8.8.8.8)	Pending	Not captured
Configure and verify Shared Clipboard (Bidirectional)	Completed	Figure 2; confirmed working by student
Configure and verify Drag-and-Drop (Bidirectional)	Completed	Figure 2; confirmed working by student
Configure and verify Shared Folder (R:\	Completed	Figure 7; confirmed working by

Task	Status	Evidence
cyberlab)		student